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Roll No

CE-5005(1)-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Water Resources Engineering

Time : Three Hours

Maximum Marks : 70

Note: i) Answer any five questions.

ii) All questions carry equal marks.

iii) Draw flow charts and diagrams, where needed.

1. a) What do you understand by Confined and unconfined Aquifer? Also explain aquiclude.
b) What do you understand by the term “Water-logging”? Explain measure to be taken to prevent water logging.
2. a) Describe various methods adopted as anti-water logging measures.
b) Derive Dupuit’s equation for steady unidirectional flow in confined aquifer.
3. a) Explain various types of cross drainage works with neat sketches.
b) Explain various types of spillways along with neat sketches.
4. a) What are the factors Affecting Duty? How can duty be improved.

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- b) An irrigation canal has gross commanded area of 80,000 hectares out of which 85% is culturable irrigable. The intensity of irrigation for Kharif season is 30% and for Rabi Season 60%. Find the discharge required at the head of the canal if the duty at its head is 800 hectares/cumecs for kharif season and 1700 hectares/cumecs for Rabi season.
5. a) Describe types of Aquifers with neat sketches.
b) Distinguish clearly between a Shallow Well and a Deep Well. How does a deep well differ from a tube well in confined aquifer?
6. a) Describe various types of irrigation with neat sketches.
b) To mature 9500 hectare of Rabi crop, the irrigation canal runs as follows:

Month	Oct	Nov	Dec	Jan	Feb
No. of days	14	28	21	21	14
Discharge (m ³ /s)	3.5	5	4.5	4.5	3.5

Calculate overall duty and depth of water.

7. Explain with neat sketch, Horton's engineering representation of 'Hydrological Cycle.'
8. Write short notes on any four of the following:
i) Salt Efflorescence
ii) Infiltration galleries
iii) Khosla's Theory
iv) Bligh's Theory
v) Specific Capacity and Specific Yield

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